

## Umelá inteligencia Zadanie 4

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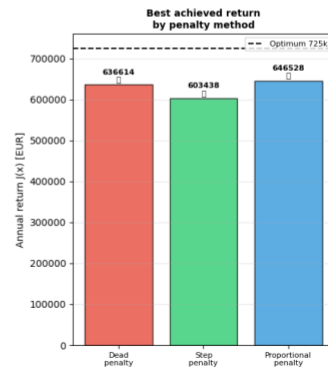
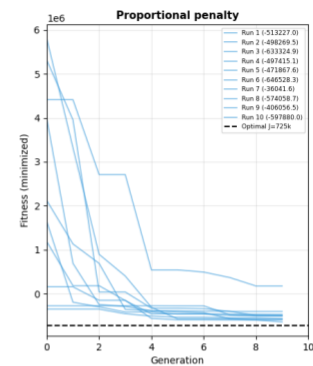
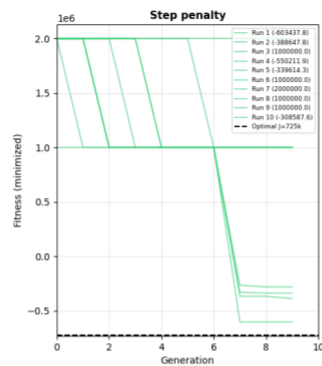
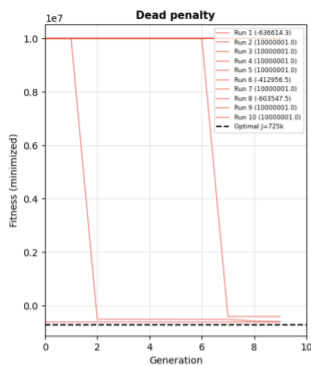
### Cieľ úlohy

Pomocou genetického algoritmu (GA) navrhnete optimálnu alokáciu investícií do vybraných finančných produktov. Podstata úlohy, vrátane metód pokutovania, bola vysvetlená v prednáške. Vyriešte príklad s rešpektovaním všetkých uvedených ohraničení. Použite pôvodný GA a samos-

tatneotestujte tri rôzne metódy pokutovania z prednášky. Pre každú metódu vykonajte výpočet zvlášť a porovnajte ich vplyv na konvergenciu fitness a na nájdené riešenie.

```
GA_PARAMS = dict(
    n_genes          = 5,
    pop_size         = 100,
    n_generations     = 10,
    mutation_rate     = 0.08,
    mutation_amp      = 500_000.0,
    crossover_pts     = 2,
    elite_count       = 5,
    lower            = 0,
    upper            = 10_000_000
)
```

Investment Allocation - A Comparison of Penalty Methods  
(GA: pop=100, gen=10, elite=5, mut=0.08)



**Dead penalty**

Best fitness (minimized): -636614.3495

Return  $J(x)$ : 636614.35 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 379449.73 EUR  |
| x2 (Featured Events)  | = | 2059457.79 EUR |
| x3 (Corporate bonds)  | = | 1934153.57 EUR |
| x4 (Government bonds) | = | 2426902.46 EUR |
| x5 (Bank savings)     | = | 2378065.49 EUR |

Constraint check:

|                                           |                          |   |
|-------------------------------------------|--------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 9178029            | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2438908                  | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=2426902, x5=2378065$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4361056 \leq 4589015$   | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                          | ✓ |
| FEASIBLE:                                 | YES                      | ✓ |

**Step penalty**

Best fitness (minimized): -603437.8305

Return  $J(x)$ : 603437.83 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 1717583.47 EUR |
| x2 (Featured Events)  | = | 765777.13 EUR  |
| x3 (Corporate bonds)  | = | 2325791.41 EUR |
| x4 (Government bonds) | = | 2090416.52 EUR |
| x5 (Bank savings)     | = | 1997360.92 EUR |

Constraint check:

|                                           |                          |   |
|-------------------------------------------|--------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 8896929            | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2483361                  | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=2090417, x5=1997361$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4416208 \leq 4448465$   | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                          | ✓ |
| FEASIBLE:                                 | YES                      | ✓ |

**Proportional penalty**

Best fitness (minimized): -646528.2815

Return  $J(x)$ : 646528.28 EUR

Allocation:

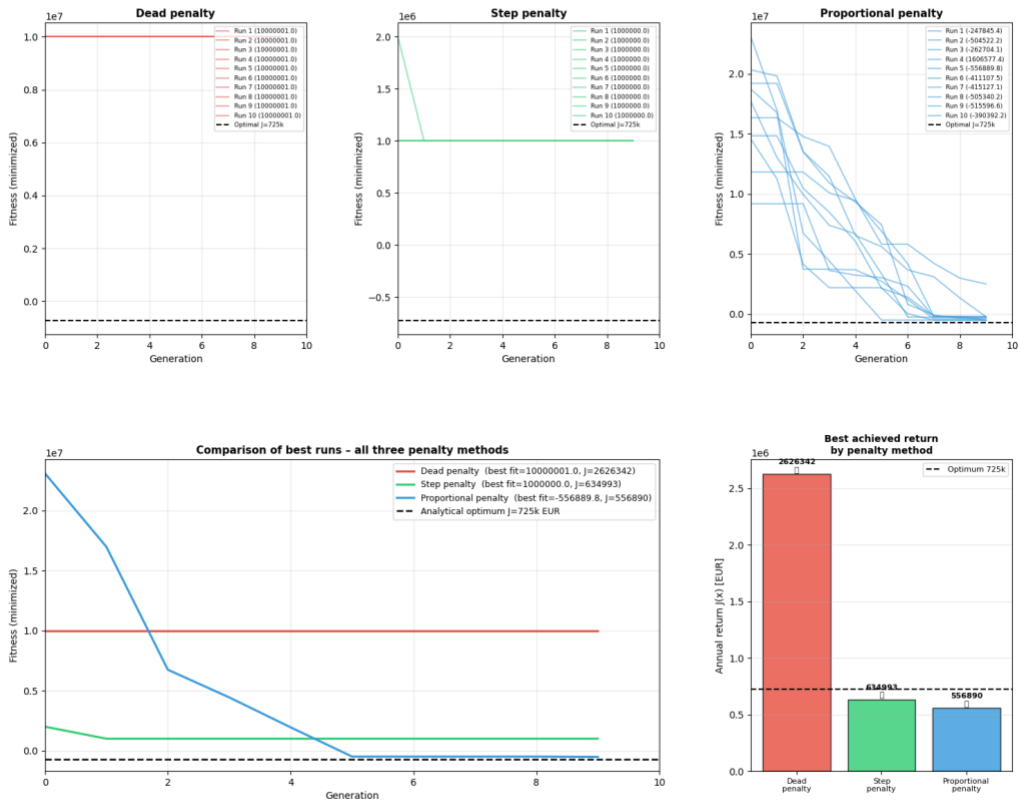
|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 1614913.67 EUR |
| x2 (Featured Events)  | = | 765779.49 EUR  |
| x3 (Corporate bonds)  | = | 2119572.41 EUR |
| x4 (Government bonds) | = | 2694282.63 EUR |
| x5 (Bank savings)     | = | 2670344.94 EUR |

Constraint check:

|                                           |                          |   |
|-------------------------------------------|--------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 9864893            | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2380693                  | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=2694283, x5=2670345$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4813855 \leq 4932447$   | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                          | ✓ |
| FEASIBLE:                                 | YES                      | ✓ |

```
GA_PARAMS = dict(
    n_genes          = 10,
    pop_size         = 150,
    n_generations    = 10,
    mutation_rate    = 0.08,
    mutation_amp     = 500_000.0,
    crossover_pts    = 2,
    elite_count      = 5,
    lower            = 0,
    upper            = 10_000_000
)
```

Investment Allocation - A Comparison of Penalty Methods  
(GA: pop=150, gen=10, elite=5, mut=0.08)



### Dead penalty

Best fitness (minimized): 10000001.0000

Return  $J(x)$ : 2626341.95 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 9008164.59 EUR |
| x2 (Featured Events)  | = | 8019464.78 EUR |
| x3 (Corporate bonds)  | = | 8447144.63 EUR |
| x4 (Government bonds) | = | 4960231.45 EUR |
| x5 (Bank savings)     | = | 9557060.68 EUR |

Constraint check:

|                                           |                          |                 |
|-------------------------------------------|--------------------------|-----------------|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 66391162           | ✗ viol=56391162 |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 17027629                 | ✗ viol=14527629 |
| P3 ( $x4 \geq x5$ ):                      | $x4=4960231, x5=9557061$ | ✗ viol=4596829  |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $13407376 \leq 33195581$ | ✓               |

P5 ( $x_i \geq 0$ ):



FEASIBLE: NO



### Step penalty

Best fitness (minimized): 1000000.0000

Return  $J(x)$ : 634992.77 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 1065061.80 EUR |
| x2 (Featured Events)  | = | 1181164.85 EUR |
| x3 (Corporate bonds)  | = | 3667186.21 EUR |
| x4 (Government bonds) | = | 1076917.53 EUR |
| x5 (Bank savings)     | = | 834064.50 EUR  |

Constraint check:

|                                           |                         |                 |
|-------------------------------------------|-------------------------|-----------------|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 39826665          | ✗ viol=29826665 |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2246227                 | ✓               |
| P3 ( $x4 \geq x5$ ):                      | $x4=1076918, x5=834064$ | ✓               |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4744104 \leq 19913332$ | ✓               |
| P5 ( $x_i \geq 0$ ):                      |                         | ✓               |

FEASIBLE: NO



### Proportional penalty

Best fitness (minimized): -556889.7805

Return  $J(x)$ : 556889.78 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 419806.96 EUR  |
| x2 (Featured Events)  | = | 1893776.17 EUR |
| x3 (Corporate bonds)  | = | 2624863.43 EUR |
| x4 (Government bonds) | = | 1356035.80 EUR |
| x5 (Bank savings)     | = | 748720.90 EUR  |

Constraint check:

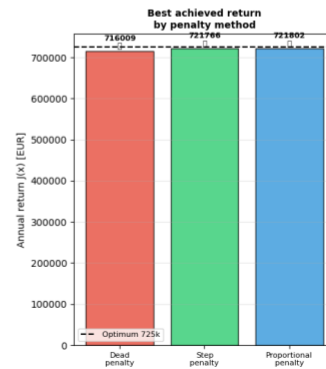
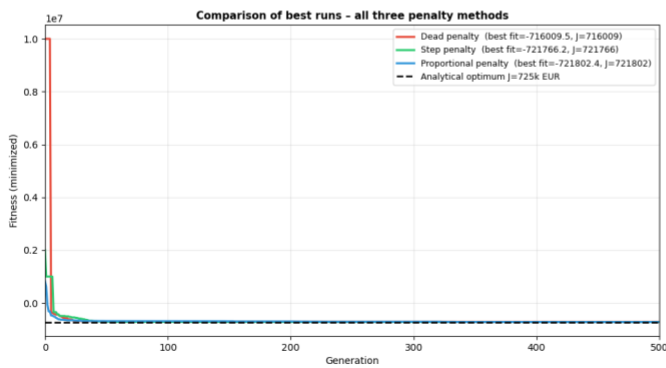
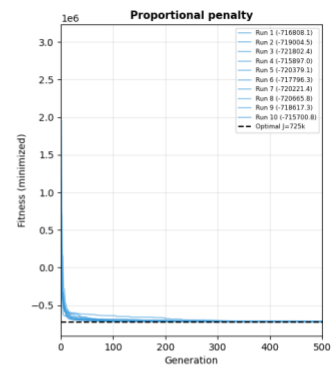
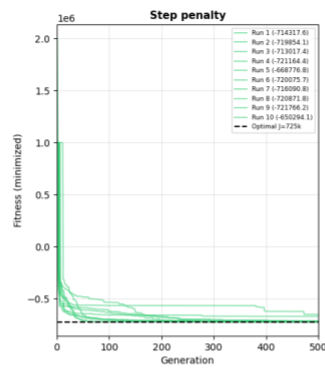
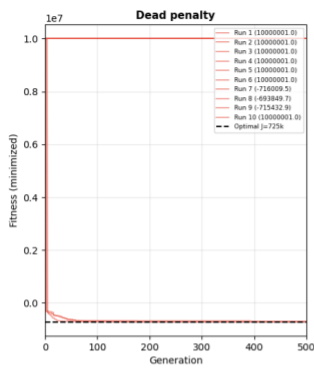
|                                           |                         |   |
|-------------------------------------------|-------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 9799278           | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2313583                 | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=1356036, x5=748721$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $3980899 \leq 4899639$  | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                         | ✓ |

FEASIBLE: YES



```
GA_PARAMS = dict(
    n_genes          = 5,
    pop_size         = 200,
    n_generations    = 500,
    mutation_rate    = 0.08,
    mutation_amp     = 500_000.0,
    crossover_pts    = 2,
    elite_count      = 10,
    lower            = 0,
    upper           = 10_000_000
)
```

**Investment Allocation - A Comparison of Penalty Methods**  
(GA: pop=200, gen=500, elite=10, mut=0.08)



### Dead penalty

Best fitness (minimized): -716009.4703

Return  $J(x)$ : 716009.47 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 0.00 EUR       |
| x2 (Featured Events)  | = | 2498924.95 EUR |
| x3 (Corporate bonds)  | = | 2329540.90 EUR |
| x4 (Government bonds) | = | 2625901.22 EUR |
| x5 (Bank savings)     | = | 2545623.03 EUR |

Constraint check:

|                                           |                             |   |
|-------------------------------------------|-----------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 9999990               | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2498925                     | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=2625901$ , $x5=2545623$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4955442 \leq 4999995$      | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                             | ✓ |
| FEASIBLE:                                 | YES                         | ✓ |

### Step penalty

Best fitness (minimized): -721766.1523

Return  $J(x)$ : 721766.15 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 0.00 EUR       |
| x2 (Featured Events)  | = | 2473666.76 EUR |
| x3 (Corporate bonds)  | = | 2446051.13 EUR |
| x4 (Government bonds) | = | 2553011.60 EUR |
| x5 (Bank savings)     | = | 2527263.18 EUR |

Constraint check:

|                                           |                             |   |
|-------------------------------------------|-----------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 9999993               | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2473667                     | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=2553012$ , $x5=2527263$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4999063 \leq 4999996$      | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                             | ✓ |
| FEASIBLE:                                 | YES                         | ✓ |

### Proportional penalty

Best fitness (minimized): -721802.4222

Return  $J(x)$ : 721802.42 EUR

Allocation:

|                       |   |                |
|-----------------------|---|----------------|
| x1 (Regular events)   | = | 0.00 EUR       |
| x2 (Featured Events)  | = | 2478262.13 EUR |
| x3 (Corporate bonds)  | = | 2446869.43 EUR |
| x4 (Government bonds) | = | 2545877.06 EUR |
| x5 (Bank savings)     | = | 2528316.24 EUR |

Constraint check:

|                                           |                             |   |
|-------------------------------------------|-----------------------------|---|
| P1 ( $\text{sum} \leq 10M$ ):             | sum = 9999325               | ✓ |
| P2 ( $x1+x2 \leq 2.5M$ ):                 | 2478262                     | ✓ |
| P3 ( $x4 \geq x5$ ):                      | $x4=2545877$ , $x5=2528316$ | ✓ |
| P4 ( $x3+x4 \leq 0.5 \cdot \text{sum}$ ): | $4992746 \leq 4999662$      | ✓ |
| P5 ( $x_i \geq 0$ ):                      |                             | ✓ |
| FEASIBLE:                                 | YES                         | ✓ |